

BORGWARNER

Final Presentation

Jason Nguyen

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About Me: Education

Lawrence Technological University

Lawrence Technological University

- ▶ Bachelor of Science in Computer Science
- ▶ Expected Graduation: May 2026
- ▶ President of The Blue Times Student Newsletter
- ▶ Social Media Manager & Varsity Player for Esports
- ▶ Team Manager for Men's Volleyball
- ▶ Research Assistant for Robotics & Perception
- ▶ Student Assistant for Python & 3D Modeling
- ▶ Teaching Assistant for Calculus Lab & R Programming Language



About Me: Experience



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BorgWarner

- ▶ IT Infrastructure Automation Intern since June 2025 under Justin Voytal
- ▶ Hosting & Cloud Engineering Team (IT Server Infrastructure, IT Storage & Backup)
- ▶ 1st Internship
- ▶ Attended the BorgWarner/Brose 2025 Car Show, Detroit Tigers vs Minnesota Twins Baseball Game, Intern Olympics 2025, Resume Revamp Workshop, and Intern Meet & Greet Event

Role in Automation

Role in Automation

- ▶ Supports the development & implementation of automating solutions to make it scalable, reliable, & efficient for IT Infrastructure
- ▶ Identify automation opportunities by developing scripts & tools to optimize the infrastructure management process
- ▶ Assist in design & development of automation scripts to streamline infrastructure operations
- ▶ Identify areas of automation & propose innovative solutions
- ▶ Support the deployment & maintenance of automated system processes
- ▶ Participate in testing & validating automation solutions to ensure they meet performance standards
- ▶ Document automation processes & provide training to team members on their use



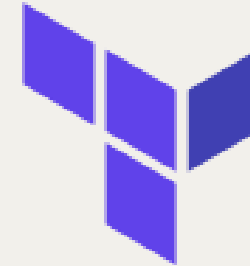
HashiCorp

Terraform

Responsibilities: Overview

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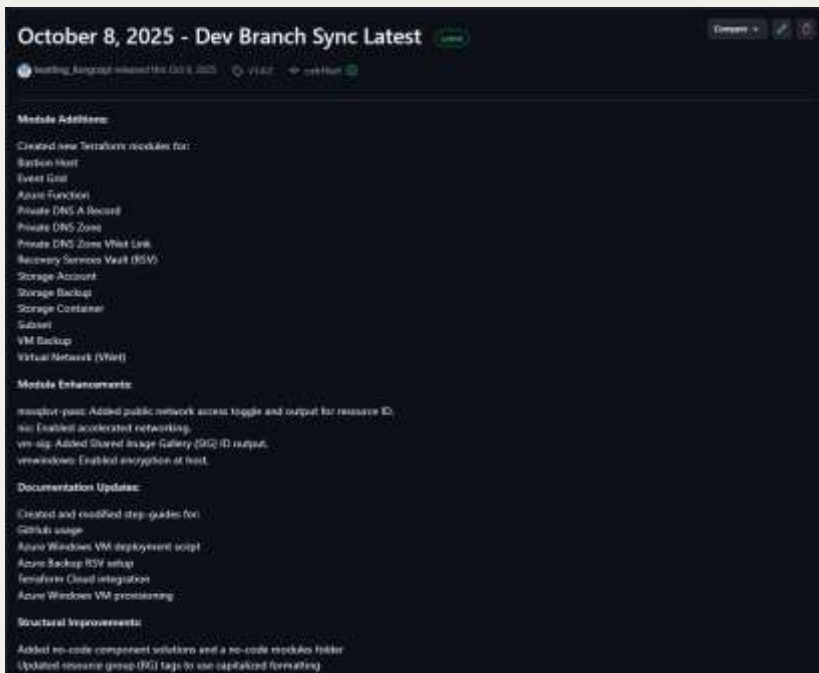
- ▶ Created Azure Backup in Terraform
- ▶ Created modules & sample code (component & solutions), coding standards, & step-guide documentation inside knowledge base Infra-KB
- ▶ Released Infra-KB version 1.4.2
- ▶ Lead Wiki.js development & authored standardized policies & configuration management database (CMDB) documentation
- ▶ Deployed infrastructure as code in Azure for MTG 350 & Atlas
- ▶ Automate creation of GitHub repositories & Terraform Cloud workspaces



Responsibilities: Deliverables

Infra-KB

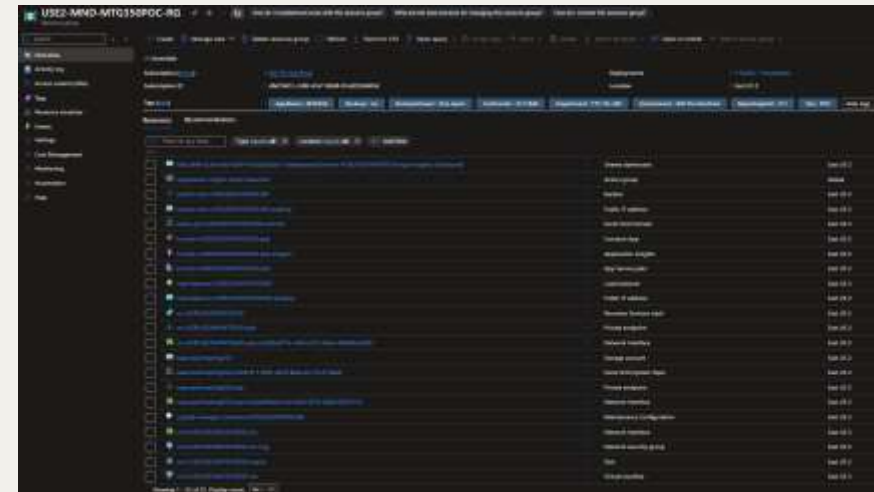
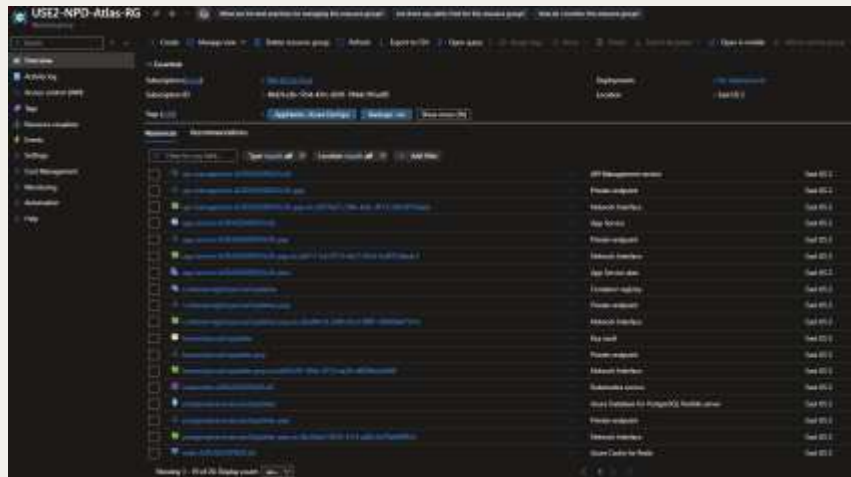
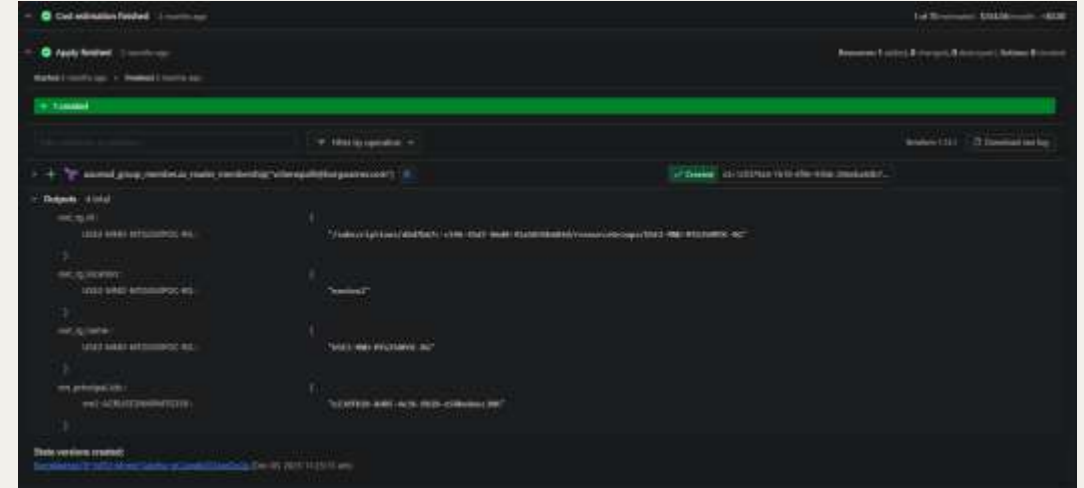
- Released Infra-KB version 1.4.2, created Azure Backup in Terraform, deployed modules & sample code (component & solutions), coding standards, & step guide documentation. Acquired new skills in Terraform creation & step-guide documentation experience.



```
1 resource "azurerm_bastion_host" "host" {
2   name           = var.bastion_host_name
3   location       = var.bastion_host_location
4   resource_group_name = var.bastion_host_resource_group_name
5
6   You, 4 months ago | 1 author (You)
7   ip_configuration {
8     name           = var.bastion_host_ip_config_name
9     subnet_id      = var.bastion_host_subnet_id
10    public_ip_address_id = azurerm_public_ip.bastion.id
11  }
12
13  depends_on = [
14    var.bastion_host_subnet_id
15  ]
16
17  You, 4 months ago | 1 author (You)
18  resource "azurerm_public_ip" "bastion" {
19    name           = var.bastion_host_public_ip_address_name
20    location       = var.bastion_host_location
21    resource_group_name = var.bastion_host_resource_group_name
22    allocation_method = var.bastion_host_public_ip_address_allocation_method
23    sku             = var.bastion_host_public_ip_address_sku
24  }
```

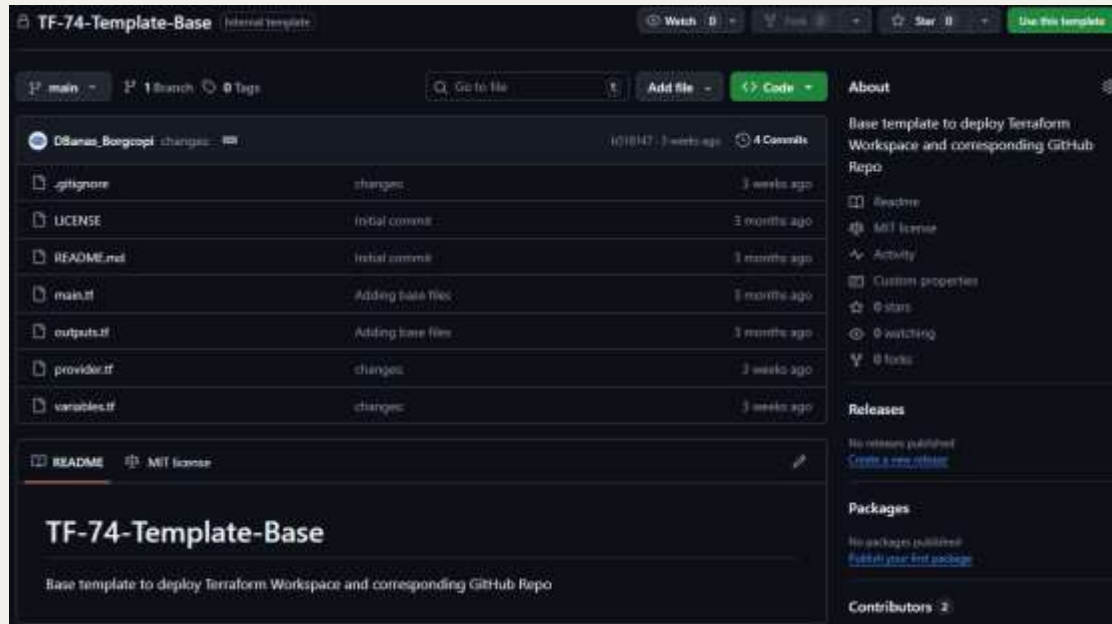
IaC Deployments

- Deployed infrastructure as code in Azure for MTG 350 & Atlas. Acquired new skills in Terraform Cloud and Azure experience.



Repository Generation

- ▶ Automate creation of GitHub repositories & Terraform Cloud workspaces using a template with a step-guide documentation. Acquired new skills in modularizing Terraform code.

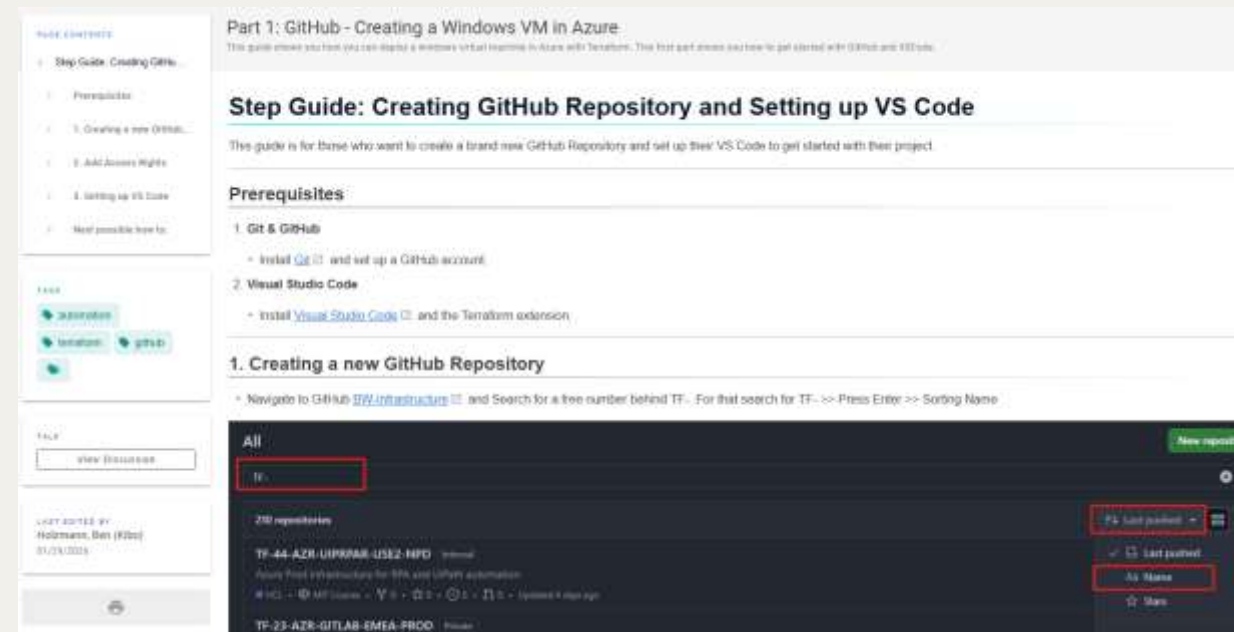
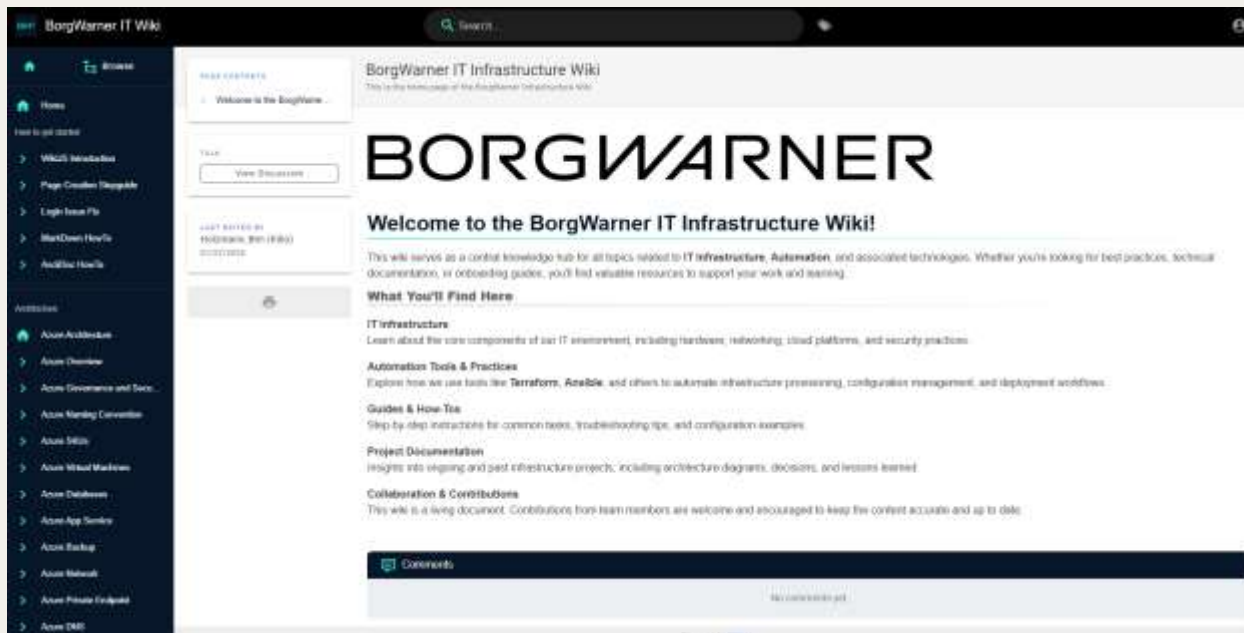


```
1 module "azure-project-repo-gen" {
2   source = "../modules/azure-project-repo-gen"
3
4   github_owner      = "BorgWarner"
5   github_template_repo = "TF-74-Template-Base" #Change to your template repo name
6   github_token      = var.github_token
7   repo_name         = "TF-75-TFPROJECTREPOGEN-test2b" #Change to your desired new repo name
8   tfe_token         = var.tfe_token
9   github_oauth_token_id = var.github_oauth_token_id
10  tfe_organization   = "BorgWarner"
11  tfe_project_id     = "prj-2xkTBVclmH92quc" #Change to your Terraform Cloud project ID, this can be found in the URL of the project in TFC
12 }
```



Wiki.js

- Lead Wiki.js development & authored standardized policies & configuration management database (CMDB) documentation. Acquired new skills in CMDB documentation.



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